

Quantum-Assisted Multi-Asset Portfolio Construction

Step 09

Portfolio Co-Pilot

Turning validated optimization results into an explainable allocation and trade plan

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1 Why a co-pilot is part of the solution

An optimization result is not useful if a decision maker cannot answer three questions: what should I hold, what must I trade, and why is the proposal allowed? Step 9 converts the validated outputs into a deterministic decision-support interface.

The co-pilot does not solve a new portfolio. It consumes the audited Step 6–7 handoff and explains it. This prevents the presentation layer from silently changing the mathematics.

2 Information architecture

The interface is organized around:

- recommended allocation and asset-class view;
- material trade tickets;
- trade-offs versus current, primary classical, and strict-warning portfolios;
- hard constraints, warning excess, and headroom;
- validation provenance; and
- predefined evidence-grounded questions.

3 Mandate-dependent recommendation

The word “recommended” requires a mandate:

- governance first: strict-warning classical;
- base objective: primary unrestricted classical;
- hybrid-method demonstration and implementation efficiency: QAOA active-rebalance.

The default co-pilot view uses the hybrid portfolio because Step 9 demonstrates the quantum-assisted workflow. It should be labeled as recommended for that purpose, not universally best.

4 From weights to dollar instructions

For portfolio value V , target weight w_i , and current weight w_i^0 ,

$$\text{target dollars}_i = Vw_i, \tag{1}$$

$$\text{trade dollars}_i = V(w_i - w_i^0). \tag{2}$$

For a \$10 million portfolio, a 5.5276 percentage-point SGOV increase is approximately

$$10,000,000 \times 0.055276 = \$552,760. \tag{3}$$

This direct mapping makes the recommendation operational.

5 Recommended hybrid allocation

Ticker	Asset class	Current	Recommended	Trade
SGOV	Cash	1.43%	6.96%	+5.53%
TIP	Inflation Linked	5.00%	5.00%	+0.00%
BWX	Core Bonds	3.98%	3.98%	+0.00%
AGG	Core Bonds	3.68%	3.68%	+0.00%
BNDX	Core Bonds	3.68%	3.68%	+0.00%
BND	Core Bonds	3.66%	3.66%	+0.00%
USMV	US Equity	3.45%	3.45%	+0.00%
TLT	Government Bonds	3.41%	3.41%	+0.00%
SHY	Government Bonds	3.40%	3.40%	+0.00%
IEF	Government Bonds	3.19%	3.19%	+0.00%
QUAL	US Equity	2.72%	2.72%	+0.00%
VTV	US Equity	2.66%	2.66%	+0.00%

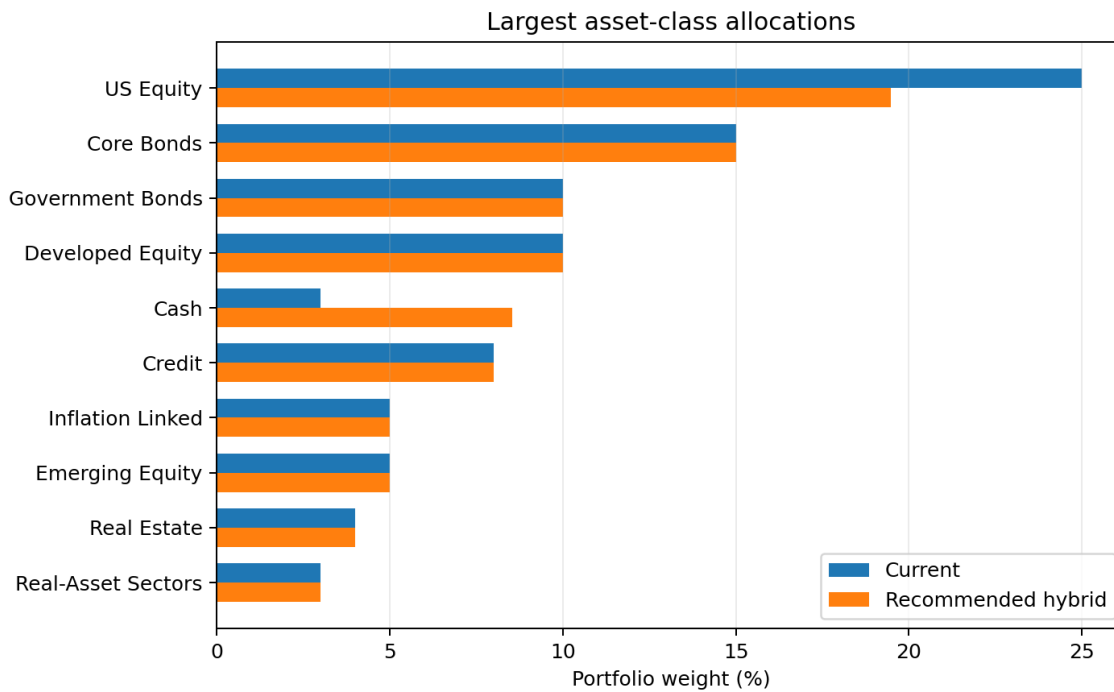


Figure 1: The hybrid rebalance reduces US equity and increases cash while leaving most sleeves unchanged.

6 Material trade tickets

The four material trades are: buy SGOV, sell MTUM, sell QQQ, and sell VUG. BIL and SHY are selected active permissions but receive no material trade.

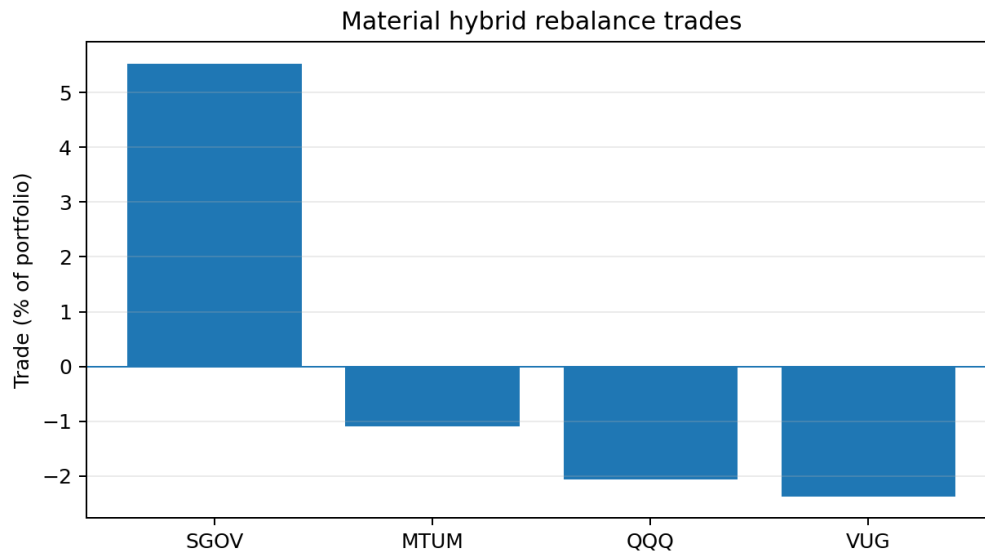


Figure 2: Material changes produced by the hybrid active-rebalance solution.

7 Trade-off explanations

Versus the current portfolio, the hybrid gives up about 0.10 percentage point of expected return while reducing volatility by about 0.95 percentage point and worst stress loss by about 1.11 percentage points. It requires 11.06% gross turnover.

Versus primary classical, hybrid expected return is about 0.04 percentage point lower, while volatility and stress loss are nearly equal. Hybrid turnover is about 2.03 percentage points lower and trading cost is slightly lower. This is the active-sleeve restriction trade-off.

Versus strict warning, hybrid uses much less turnover and cost, but strict warning has lower volatility, lower stress loss, no soft warning breaches, and better forward downside behavior.

8 Constraint evidence

The co-pilot states each rule as value, limit, and status. It separates the Step 6 policy audit from the larger Step 7 lifted-QP audit. A user sees not only “PASS” but also how close the portfolio is to a boundary.

9 Scenario warning and hard-limit headroom

Scenario	Loss	Warning excess	Hard headroom
Global equity selloff	11.4143%	0.0042%	3.9958%
Inflation / rate shock	7.6814%	0.7096%	3.2904%
Credit / liquidity crisis	10.4030%	0.4940%	3.5060%
Commodity supply shock	3.0750%	1.0467%	2.9533%
Broad deleveraging	12.7160%	0.3890%	3.6110%

All hard limits pass. All five warnings are exceeded, though the global equity warning is exceeded by only 0.0042 percentage point. The proper description is hard-limit compliant with five soft-warning breaches.

10 Deterministic question answering

The interface answers predefined questions such as:

- Why not choose primary classical?
- Why not choose strict warning?
- Which constraint is closest?
- How much return is sacrificed for risk reduction?
- Why are six assets selected but only four traded?
- Does this prove quantum advantage?

Answers are assembled from stored metrics and validation evidence. This prevents unsupported narrative claims.

11 Provenance and auditability

A **technical** explanation should identify data source, cost scenario, policy mode, solver sources, QAOA configuration, exact-enumeration status, and evidence tier. The co-pilot therefore acts as a view over the audit trail, not a black-box chatbot.

12 Reproduction checklist

Load the same portfolio snapshot, verify that weights sum to one, recompute trade deltas and dollar values, reconcile asset-class totals, compare all metric deltas, validate constraint statuses, and test every narrative against its underlying data field.

13 What Step 9 establishes

Step 9 demonstrates that the project can communicate an allocation, implementation plan, trade-offs, and policy evidence to a non-technical decision maker. It does not add new empirical evidence or change the underlying winner.

A Why a co-pilot is a decision layer, not another optimizer

The co-pilot does not change weights or rerun the mathematical model. It consumes validated outputs and translates them into decisions a reviewer can inspect: recommended allocation, dollar trades, comparison with a baseline, hard-policy status, warning headroom, method evidence, and limitations. Keeping calculation and presentation separate prevents a user-interface change from silently altering the optimization result.

B Data snapshot contract

The application is driven by a deterministic snapshot containing:

- all 50 current and recommended weights;
- ticker and asset-class labels;
- portfolio metrics for current, primary, strict, and QAOA candidates;
- material trades and selected active support;
- scenario warning and hard-limit data;
- Step 6 and Step 7 evidence counts;
- QAOA configuration and exact-enumeration verdict; and
- run scope and claim limitations.

Weights must sum to one and displayed differences must recompute from the snapshot. The interface must not contain manually typed metric values that can drift from the exported evidence.

C Dollar-allocation conversion

For portfolio value V , dollar allocation is

$$X_i = Vw_i. \quad (4)$$

Dollar trade is

$$\Delta X_i = V(w_i - w_i^0). \quad (5)$$

For a \$10 million portfolio, a 5.5276 percentage-point SGOV increase corresponds to roughly

$$10,000,000 \times 0.055276 \approx \$552,760. \quad (6)$$

The same calculation is applied interactively when the user changes portfolio value.

D Allocation story

At asset-class level, the QAOA portfolio reduces US equity from approximately 25.00% to 19.4724% and raises cash from 3.00% to 8.5276%, while other asset-class exposures remain unchanged. This is consistent with the four material trades: buy SGOV and sell MTUM, QQQ, and VUG.

The interface must distinguish selected permissions from executed trades. Six assets belong to the selected support, but only four have material weight changes. BIL and SHY are valid support coordinates with zero material final trade.

E Trade-off calculations

For a metric m , comparison with baseline b is

$$\Delta m = m_{\text{recommended}} - m_b. \quad (7)$$

The unit must remain explicit. A difference between 7.87% and 8.82% is -0.95 percentage point, not -0.95% relative change. The co-pilot uses paired values and written interpretations so a reviewer sees both the level and the difference.

Relative to current, QAOA lowers expected return modestly while reducing modeled volatility and worst stress. Relative to primary classical, it has nearly identical modeled risk but lower turnover and cost. Relative to strict warning, it trades less but retains five soft warnings and somewhat higher downside risk.

F Constraint language

The correct statement for QAOA is

Hard-limit compliant under the declared soft-warning policy, with five soft-warning breaches.

The interface displays warning excess and hard-limit headroom separately. For scenario s ,

$$\text{warning excess}_s = \max(L_s - \tau_s, 0), \quad (8)$$

$$\text{hard headroom}_s = H_s - L_s. \quad (9)$$

A positive warning excess and positive hard headroom are not contradictory.

G Evidence hierarchy

The co-pilot distinguishes:

- Step 6 portfolio-policy audit rows;
- Step 7 lifted-QP constraint rows;
- selected independent solver and KKT status;
- exact QUBO enumeration over 462 states;
- seed-level QAOA reliability; and

- synthetic forward-path scope.

These counts answer different questions and must not be merged into one ambiguous “constraints passed” number.

H Deterministic question answering

The question panel uses a fixed library of evidence-grounded answers. Questions include why QAOA was selected for demonstration, why strict warning may be preferable for governance, why six selected coordinates produce four trades, what the closest scenario limit is, and whether quantum advantage is claimed. A deterministic lookup is safer than unconstrained generative text for a **audited prototype** because every answer can be traced to data.

I User-interface validation

Automated tests should verify weight sums, dollar conversions, trade signs, asset-class aggregation, metric deltas, scenario formulas, candidate labels, evidence counts, and prohibited claims. Visual tests should check responsive layout, readable charts, no clipped labels, and accessibility of table values without relying on color alone.

J Reproduction protocol

1. Export one validated JSON snapshot from Steps 6 and 7.
2. Recompute all derived fields in a build script.
3. Render allocation, trades, trade-offs, constraints, and evidence tabs from that snapshot.
4. Test every displayed number against the source data.
5. Serve the application locally or through GitHub Pages.
6. Confirm the interface still works without external APIs.
7. Preserve the synthetic-model and no-quantum-advantage disclosures on every relevant view.

Interface and terminology clarification

These notes align interpretation with the final executed code path and saved evidence without changing the reported numerical results.

- The co-pilot is a deterministic view over saved evidence. It does not run a learned model or create a new investment recommendation independently of Steps 3-7.
- References to the "current portfolio" in the interface mean the synthetic/model reference incumbent used by the experiment.
- The interface distinguishes selected permissions from material executed trades and reports provenance/validation rather than generating unsupported explanations.